

Roll No.

TBT-301

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

GENETICS AND MOLECULAR BIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the Mendel's law of inheritance and their significance. Why is law of dominance not widely accepted ? (CO1)

OR

- (b) What will happen when mother with recessive allele (dd) for D-antigen is conceived with fetus carrying dominant (DD) allele ? Which Mendel's law prevails here ? (CO1)

2. (a) Discuss the characteristics of dominant and recessive type sex-linked inheritance with an example. (CO1)

OR

- (b) Many genetic diseases are caused by error in cell division. What do you think could have happened to chromosomes during meiosis ? Name few types of human abnormalities because of chromosomal aberrations.

(CO1)

P. T. O.

(2)

3. (a) Define Recombinant frequency. Find the R. F., if the following traits are observed in *Drosophila* : red eye normal wing = 368, red eye vestigial wing = 56, white eye vestigial wing = 336, white eye normal wing = 40. (CO1)

OR

- (b) Explain Chromosomal theory of linkage proposed by Morgan. (CO2)

4. (a) What do you know about Hardy-Weinberg equilibrium ? What are the rules for this to be in equilibrium ? (CO1)

OR

- (b) If the genotype of 5 individual is AA, AA, Aa, Aa, aa. If $p + q = 1$, then what will be (i) Allele frequency of dominant and recessive (ii) Homozygous dominant, homozygous recessive, heterozygous dominant ? (CO2)

5. (a) Draw structures of all purine and pyrimidine residues. (CO2)

OR

- (b) Illustrate nucleosome model and how it justifies the packaging of long DNA in the tiny invisible nucleus. (CO2)

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TBT-302

**B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

CELL BIOLOGY

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by light microscopy ? Write its types and discuss any *one* in detail. (CO1/CO2)

OR

- (b) What is electron microscopy ? How does TEM differ from SEM ? (CO1/CO2)

2. (a) Write down *five* similarities and *five* differences in Eukaryotes and Prokaryotes. (CO1/CO2)

OR

- (b) What is endosymbiont theory ? Write the name of eukaryotic cell organelles evolved through his theory. (CO1/CO2)

P. T. O.

(2)

3. (a) Write down *five* similarities and *five* differences between Plant and Animal Cells. (CO1/CO2)

OR

- (b) Write short notes on the following : (CO1/CO2)

(i) Plant Cell Wall

(ii) Chloroplast

4. (a) Write down the process of electron transport chain. (CO1/CO2)

OR

- (b) Write short notes on the following : (CO1/CO2)

(i) Glycosidic bond

(ii) Peptide bond

5. (a) Discuss why mitochondria and chloroplast are called power generators of the cell. (CO2)

OR

- (b) Explain how DNA is packaged into chromosome inside a eukaryotic cell. (CO2)

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TBT-303

B. TECH. (BIOTECHNOLOGY) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MICROBIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Give the composition of eubacteria cell wall using suitable diagram.

(CO1/CO2)

OR

- (b) Explain how the bacterial cell wall differs in Gram positive and Gram negative bacteria. Give suitable diagram.

(CO1/CO2)

2. (a) Using suitable diagram give the shape and arrangements rod shaped bacteria. Name a method of its identification in lab.

(CO1)

OR

- (b) What is the significance of negative staining in bacterial shape identification ? Briefly give the method of staining. Make suitable diagram.

(CO1)

P. T. O.

(2)

3. (a) Discuss the function of plasmid found in bacteria. (CO1)

OR

- (b) List the different essential organelles present in the prokaryotic cell. Explain the function of any *one*. (CO1)

4. (a) What are the gene transfer mechanism found in prokaryotes ? Discuss any *one*. (CO1)

OR

- (b) A test-tube containing *Candida*, staphylococci, streptococci, diplococci cells is provided to you. How it can be identified by staining and differentiated under microscope ? Give a brief process. (CO1)

5. (a) What do you mean by biogenesis theory and abiogenesis theory ? (CO2)

OR

- (b) What do you mean by Golden Age of Microbiology ? Give the contribution of Robert Koch in Microbiology. (CO2)